

Adversarial Attacks & Defences

Tesla vehicles have been fooled by targeted stickers. Facial-recognition by adversarial glasses. Fraud detection by word-level attacks on complaint text. Adversarial ML isn't theoretical — it's happening today, at scale, in production.

When was your model last adversarially tested? Honestly.

WHAT'S INSIDE

- 01** Three attack classes
- 02** FGSM — simplest adversarial
- 03** 5 defences
- 04** Prompt injection — LLM-specific
- 05** Red-team your own model

WHY THIS LESSON MATTERS

A sticker on a stop sign. A car crashes at 80 km/h.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

- 1 Classify attacks: evasion, poisoning, extraction.
- 2 Demonstrate FGSM attack.
- 3 Apply 5 defence strategies.
- 4 Detect prompt injection.
- 5 Red-team your own model.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Three attack classes

Key concept of this lesson.

02

FGSM — simplest adversarial

Fast Gradient Sign Method

03

5 defences

Layered

04

Prompt injection —...

The 2026 defining attack

01

PART 1 OF 4

Three attack classes

Key concept of this lesson.

Three attack classes

A core idea you'll use throughout this lesson.

DEFINITION

Three attack classes

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

FGSM — simplest adversarial

Compute gradient of loss wrt input.

Take small step along its sign.

Input looks visually identical.

Model output wildly different.

DEMO

8 lines of PyTorch

Show in class. Room audibly gasps.

KEY POINTS

- Compute gradient of loss wrt input
- Take small step along its sign
- Input looks visually identical
- Model output wildly different

CONCEPT 3 OF 4

5 defences

Adversarial training — include adv examples.

Input validation — bound + sanity-check.

Detection — auxiliary classifier for adv inputs.

Differential privacy — noise during training.

Watermarking — prove ownership if model stolen.

DEPTH

Layers > single fence

No single defence stops all attacks.

DEFINITION

5 defences

Layered

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Prompt injection — LLM-specific

Treat all external input as untrusted.

Structure outputs (JSON schema); validate.

Log inputs + outputs for audit.

Least-privilege tools.

HITL for destructive actions.

LLM AGE

Prompt injection

Covered W5.7; repeat here for security framing.

KEY POINTS

- Treat all external input as untrusted
- Structure outputs (JSON schema); validate
- Log inputs + outputs for audit
- Least-privilege tools

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Break a classmate's model

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

- 1
- 2 Swap models.
- 3
- 4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which attack class most threatens your model?

Q2

Is adversarial training in your training loop?

Q3

Quarterly red-team on your roadmap?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Three attack classes:

A

Speed, space, cost

B

Evasion, poisoning, extraction

C

UI, backend, DB

D

None

Three attack classes:

CORRECT ANSWER



Evasion, poisoning, extraction

WHY

Canonical taxonomy

FGSM:

A

A defence

B

An attack

C

A framework

D

None

FGSM:



CORRECT ANSWER

An attack

WHY

Fast Gradient Sign Method — evasion attack

Adversarial training:

A

Slower inference

B

Include adv examples in train

C

Prohibited

D

Hardware

Adversarial training:

CORRECT ANSWER



Include adv examples in train

WHY

Most common defence

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

A team applies the lesson's main idea to a real workflow — and the results show up within a quarter.

THE TAKEAWAY

"The paranoia is the feature."

WHAT TO NOTICE

1

Three attack classes

2

FGSM — simplest adversarial

3

5 defences

4

Prompt injection —...

FGSM + mitigation

On your W7 image classifier.

DELIVERABLES

- Run FGSM.
- Document adversarial examples.
- Apply adversarial training.
- Re-test.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Before/after table.

ONE THING TO REMEMBER

“

"The paranoia is the feature."

— AI Learning Hub

END OF LESSON 8

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 5 — Portfolio.